

Predicting the Zero-Shot Transfer of a Promptable Video Tracker

Can we forecast SAM 3’s per-species, per-place reliability — for detection and association separately — from label-free distances?

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Abstract

Promptable trackers such as SAM 3 [1] segment and track any named object in a video with no training on that category. Conservation teams now point them at new species in new places constantly — but nobody can say, *in advance*, whether the model is trustworthy for a given animal in a given location. This project builds and validates a **predictive rule** for that reliability. Using SA-FARI [2] — whose train/test splits are disjoint by *species and location* — we measure SAM 3’s zero-shot promptable-tracking accuracy with the *official* evaluator, decompose it into **detection** (pDetA) and **association** (pAssA) via the HOTA family [3], and ask whether each can be predicted from four **label-free distances** (taxonomic, visual, environment, temporal) computed against the training split alone. The heavy model is frozen throughout; only a small statistical model is fit. Predictable out-of-distribution generalisation is established for *classifiers* [4, 5] but not for trackers; our contribution is the first such study for a *promptable video tracker*, decomposed into two predictors, yielding a deployment “will it work here?” estimator with honest confidence intervals. The result is informative either way.

1 Background and motivation

A **promptable tracker** is a vision-language model that, given a text prompt like “impala”, segments and tracks every matching animal across a video with no training on that species (zero-shot). SAM 3 [1] is the current exemplar. This capability is transforming camera-trap ecology: a single model can be aimed at any species anywhere. But it creates a new, unanswered question — *when should we trust it?* Teams deploy these models on unseen species and regions with no rule that predicts, before running, whether the output is reliable. That gap is what we fill.

The idea that generalisation can be *predicted* is not new for classifiers. “Accuracy on the Line” [4] showed that, across models and shifts, out-of-distribution accuracy correlates strongly and linearly with in-distribution accuracy; “Agreement on the Line” [5] made the estimate *label-free* by using inter-model agreement. Both concern classification. Two further results motivate *which* properties should predict transfer here: vision foundation models carry taxonomic structure across the tree of life (BioCLIP [6]), motivating taxonomic/visual distance; and camera-trap recognition degrades sharply at new locations (Terra Incognita [7]), motivating an environment distance. What is missing is any predictable-transfer study for a *promptable video tracker*, and any treatment of the fact that tracking has two distinct failure modes — failing to *find* the animal, and failing to *follow* it — that may be governed by different factors.

The metric makes that decomposition clean. Promptable HOTA (pHOTA) factors, HOTA-style [3], into pDetA (detection: did it find the animals?) and pAssA (association: did it keep each identity over time?). Analysing the two *separately* is the intellectual core of this proposal: we hypothesise they transfer for different reasons.

SA-FARI is a natural experiment. The largest open multi-animal wildlife tracking dataset (Meta $\times$ Conservation X Labs) [2] spans **99 species** with a full 7-level taxonomy, **741 locations** across four continents, and a decade of footage. Crucially, its train and test splits are **disjoint by species and location**, so every test video is a controlled probe of transfer along a *measurable* distance from the training set.

2 Research question and hypotheses

RQ. Can we predict SAM3’s zero-shot **pDetA** and **pAssA** on an unseen (species, place) *before running it*, from properties measurable in advance — and which factors govern transfer, separately for finding the animal and following it?

H1 (detection)

pDetA falls with **species novelty** — taxonomic and visual distance from the nearest training species.

H2 (association)

pAssA falls with **environment difficulty** — scene distance, day/night/IR, clutter, camera motion — largely regardless of species.

H0 (null, still a result)

the distances explain little variance $\Rightarrow$ transfer is not simply distance-driven; pivot to representational probing (does SAM3 encode taxonomy?).

Success is a *validated, out-of-sample* predictor of **pDetA/pAssA** from label-free distances, with honest confidence intervals — *not* a higher tracking score.

3 Proposed approach

Measurement. We run **frozen** SAM3 over the SA-FARI test split (species-specific prompts as the primary condition; a generic “animal” prompt for robustness), and score with the **official VEval** evaluator — *the metric is never re-implemented*. Scores are aggregated to a **cell** = a (species, location_id, time) group, carrying *support* counts (n_{frames} , n_{masklets} , n_{videos}). Hard negatives (a query returning nothing) are kept — they drive false-positive analysis.

Four label-free distances (all computed against the *frozen* training set, using no label of the target species): **taxonomic** (tree steps to the nearest training species via the lowest common ancestor); **visual** (cosine distance of DINOv2 [8] embeddings of the *mask-cropped* animal to the nearest training-species prototype); **environment** (scene embedding of the animal-masked-out background to the nearest training location, plus a night/IR flag); and **temporal** (years to the nearest relevant training footage). A **SAM-3 familiarity proxy** — separability of the test species in SAM3’s own feature space — hedges against the fact that SAM3’s true pretraining corpus is unknown; a CLIP [9] variant of the visual distance is a robustness check.

Fitting the law. We model each bounded target with **beta regression / logit-link GLM**, *weighted by support* and including a $\log(n_{\text{frames}})$ covariate so that “rare” is never mistaken for “far”. Variance is attributed with dominance/commonality analysis (not four univariate fits) with collinearity (VIF) controlled. The predictor is then validated **out-of-sample** by holding out *whole species and whole locations* in cross-validation, and packaged as a reliability estimator: four distances in $\rightarrow$ predicted **pDetA/pAssA** + interval out.

4 Novelty (stated honestly)

The contribution is *not* that OOD performance can be predicted (established for classifiers [4, 5]), nor that embeddings carry taxonomy [6], nor that camera-trap models shift by location [7]. It is

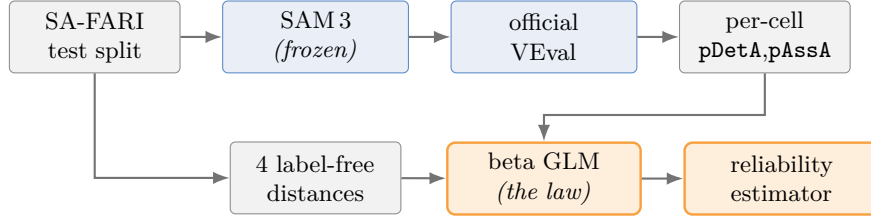


Figure 1: Pipeline. SAM3 and VEval are used off-the-shelf (SAM3 frozen, the metric official); only the small *beta GLM* is fit, from label-free distances, and packaged as a deployment reliability estimator. Nothing about the target species’ labels enters a distance.

the *combination*: to our knowledge, the **first predictable-transfer study for a promptable video tracker, decomposed into two predictors** (detection vs association, via HOTA), on SA-FARI’s species $\times$ place $\times$ time natural experiment — yielding a practical *deployment reliability estimator*.

	Classifier OOD line [4, 5]	This project
Model studied	image classifiers	a <i>promptable video tracker</i> (SAM3)
Predicted quantity	one accuracy	<i>two</i> targets: detection & association
Predictor	in-distribution acc. / agreement	four <i>label-free distances</i> of the probe
Output	a correlation	a validated per-(species,place) reliability tool

Table 1: Where the study sits relative to the established OOD-predictability line.

5 Evaluation, success criterion and constraints

Out-of-sample validation. The headline is not a within-sample fit but a *predictor*: leave-one-species-out and leave-one-location-out cross-validation (whole groups held out) predict pDetA/pAssA from distances alone, reported with bootstrap confidence intervals and calibration. Two decision gates keep the project honest: (i) after measurement, are the scores in SA-FARI’s SAM3 ballpark? (ii) after validation, is there real out-of-sample signal — if not, invoke H0 and pivot to representational probing.

Hard constraints. Freeze the seen set *first* and compute every distance against the training split only (no test information may enter a feature — leakage would make the “prediction” trivial). Use the *official* evaluator for all metrics. Model bounded scores with support-weighted beta/logit GLMs. Hold out whole species and locations. Keep hard negatives. Inference-only: a single modern GPU suffices.

Indicative timeline

- Wk 1** Setup; acquire SA-FARI; freeze the seen/unseen reference (leakage firewall).
- Wk 2–4** Frozen SAM3 inference over the test split; score with VEval; per-cell pDetA/pAssA (gate 1).
- Wk 4–6** Build the four label-free distances + familiarity proxy; assemble the feature table (leakage test).
- Wk 6–9** Fit per-target beta GLMs; variance partitioning; the detection/association decomposition test.
- Wk 9–12** Grouped CV (leave-species/location-out); bootstrap CIs; predictive-line figures (gate 2).
- Wk 11–13** Factor ablations and design/confound robustness (encoder, crop, prompt, aggregation).
- Wk 13–14** The reliability estimator; dissertation write-up.

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